

Training Configuration and Summary:

Dataset used: rssi_odom_dataset_11apr25_ajust.parquet

LSTM Multi-Step Training Report

Report Generated: 2025-05-08 10:26:35

Training Start: 2025-05-08 10-14-14

Training End: 2025-05-08 10-22-04

Training Duration: 7.83 minutes

Best Hyperparameters:

Number of Layers: 1

Activation: relu

Optimizer: Nadam

Dropout Rate: 0.20

MAE: 2.1676

MSE: 6.8173

RMSE: 2.6110

R2 Score: -0.1659

Multi-Step Forecast:

Forecast Steps: 20

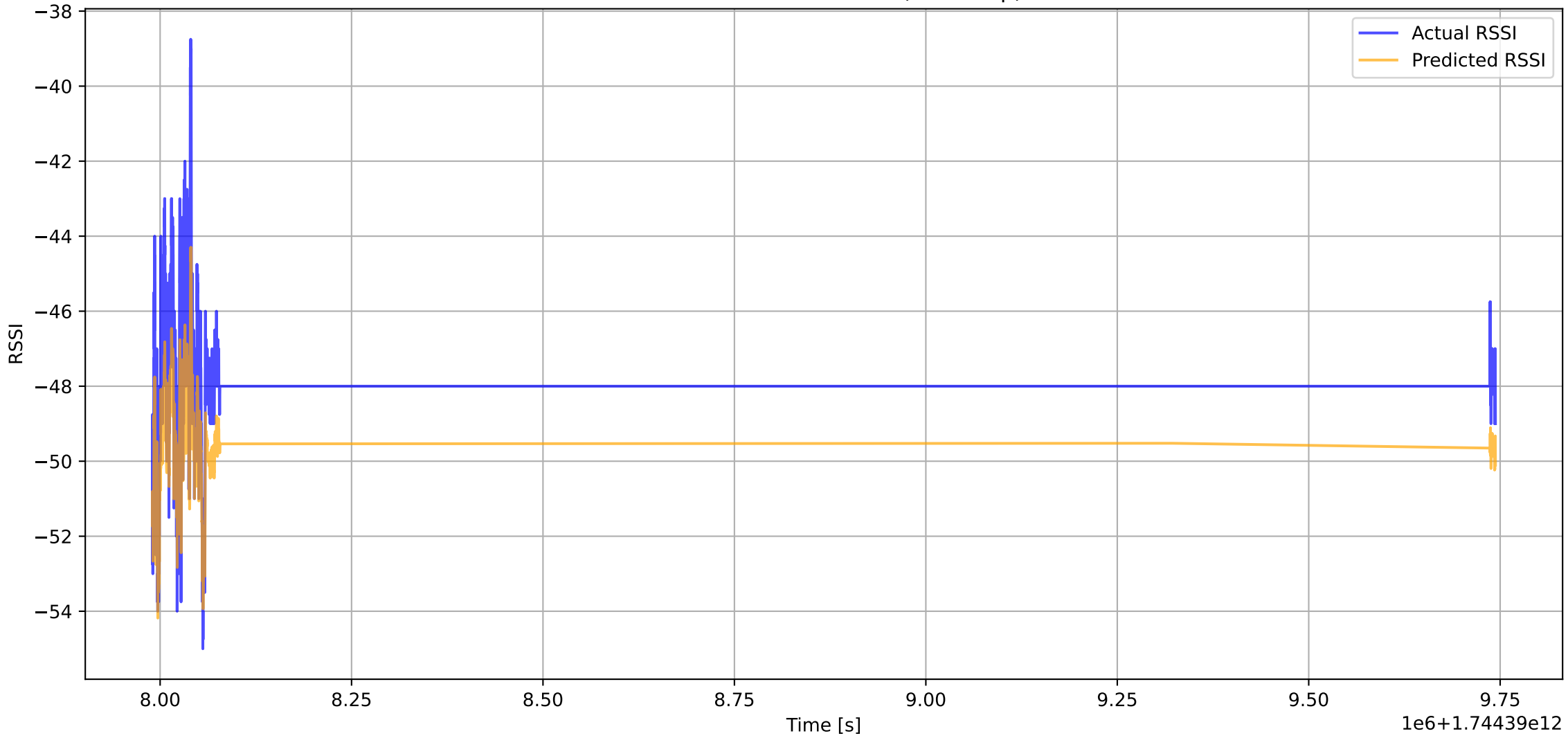
MAE: 0.0670

MSE: 0.0058

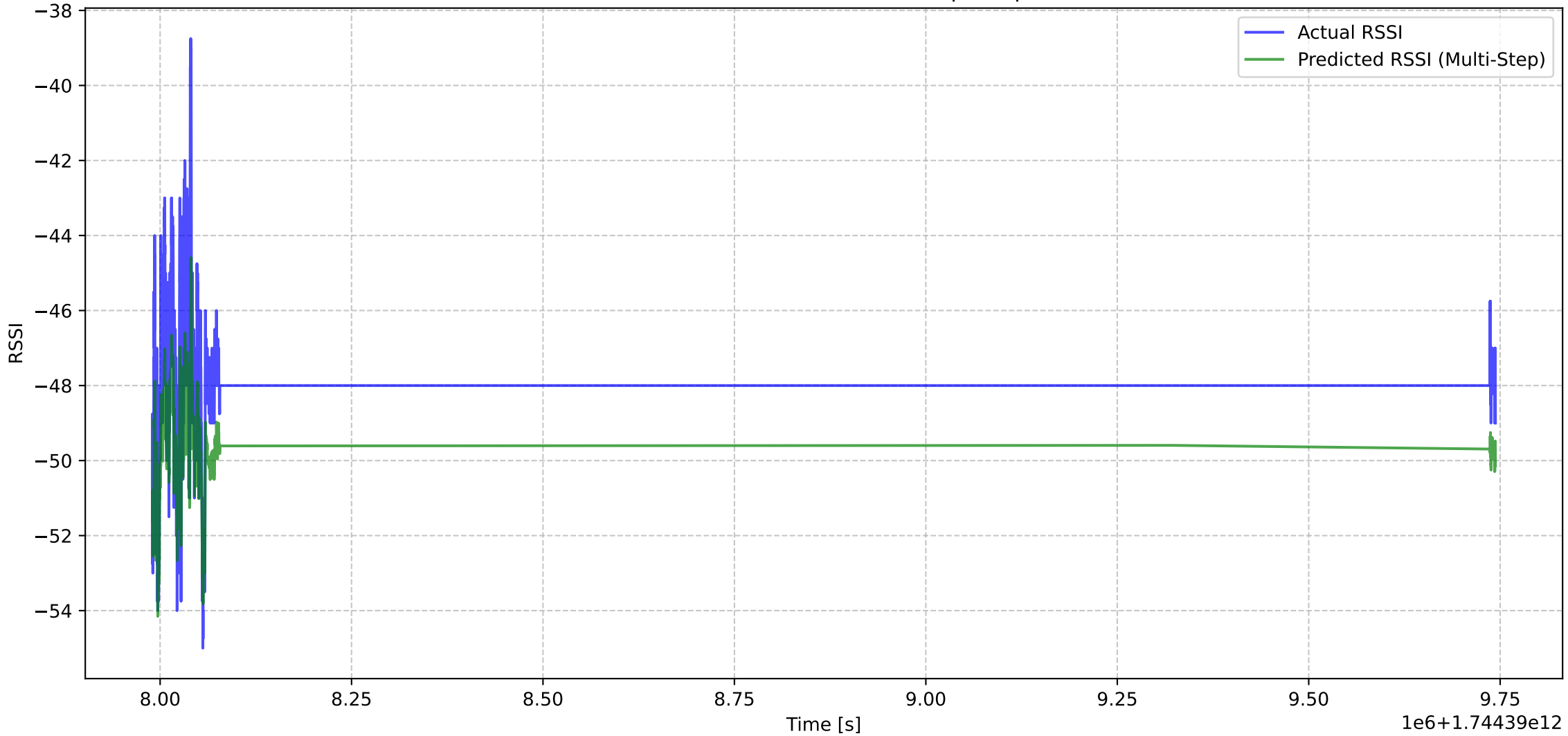
RMSE: 0.0763

R2 Score: 0.2202

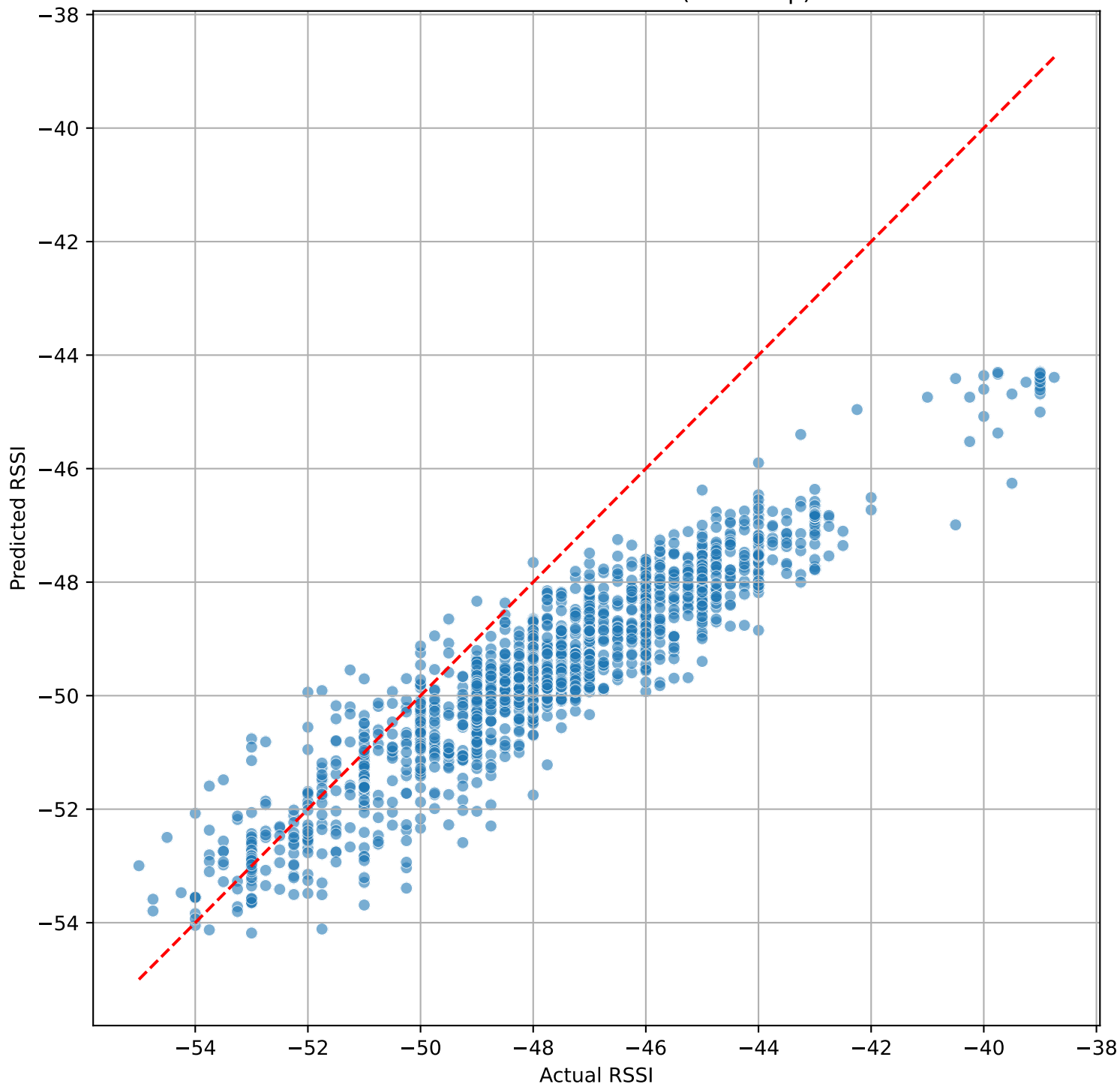
Actual vs Predicted RSSI (First Step)



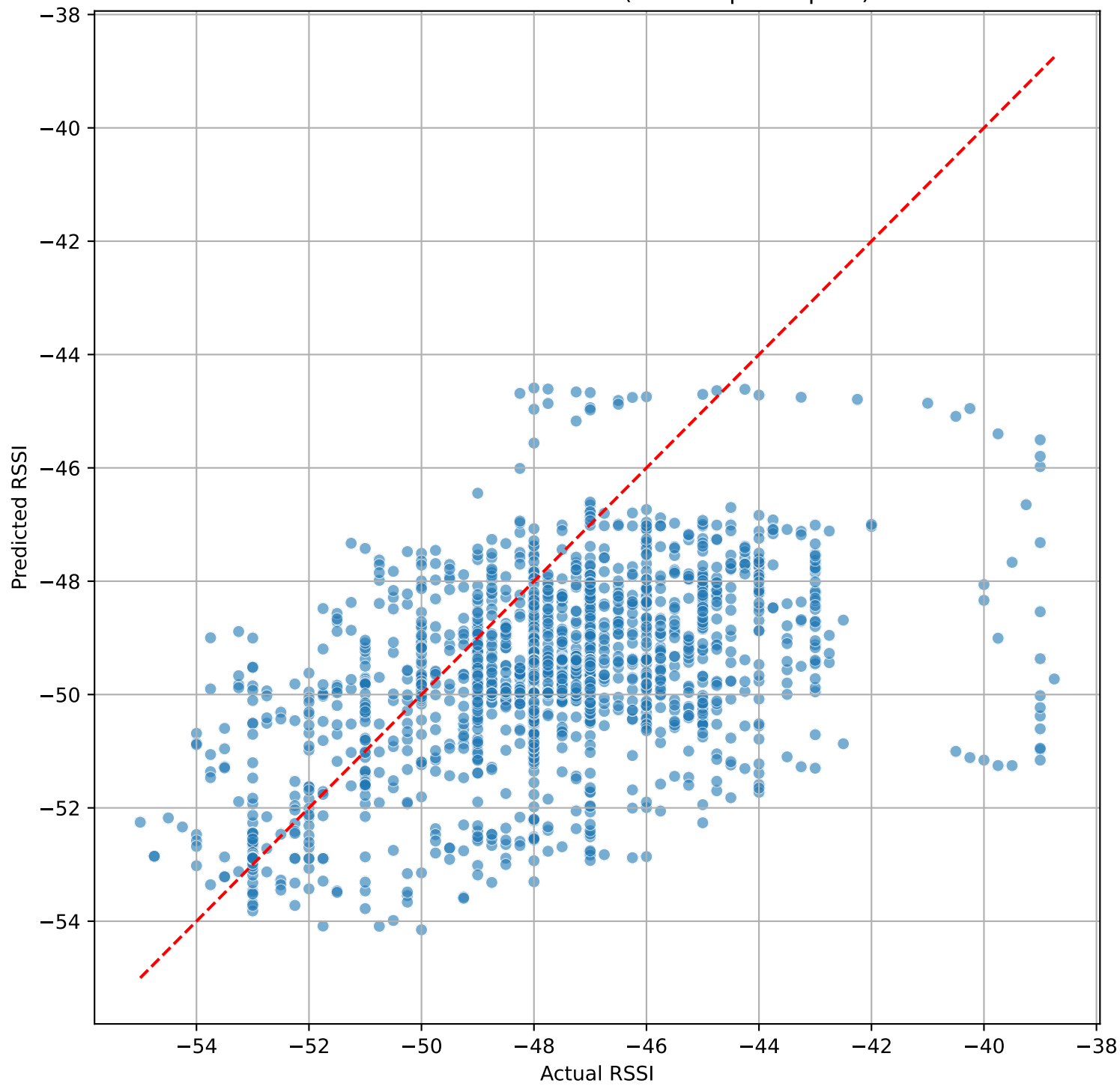
Actual vs Predicted RSSI (Multi-Step (step 20))



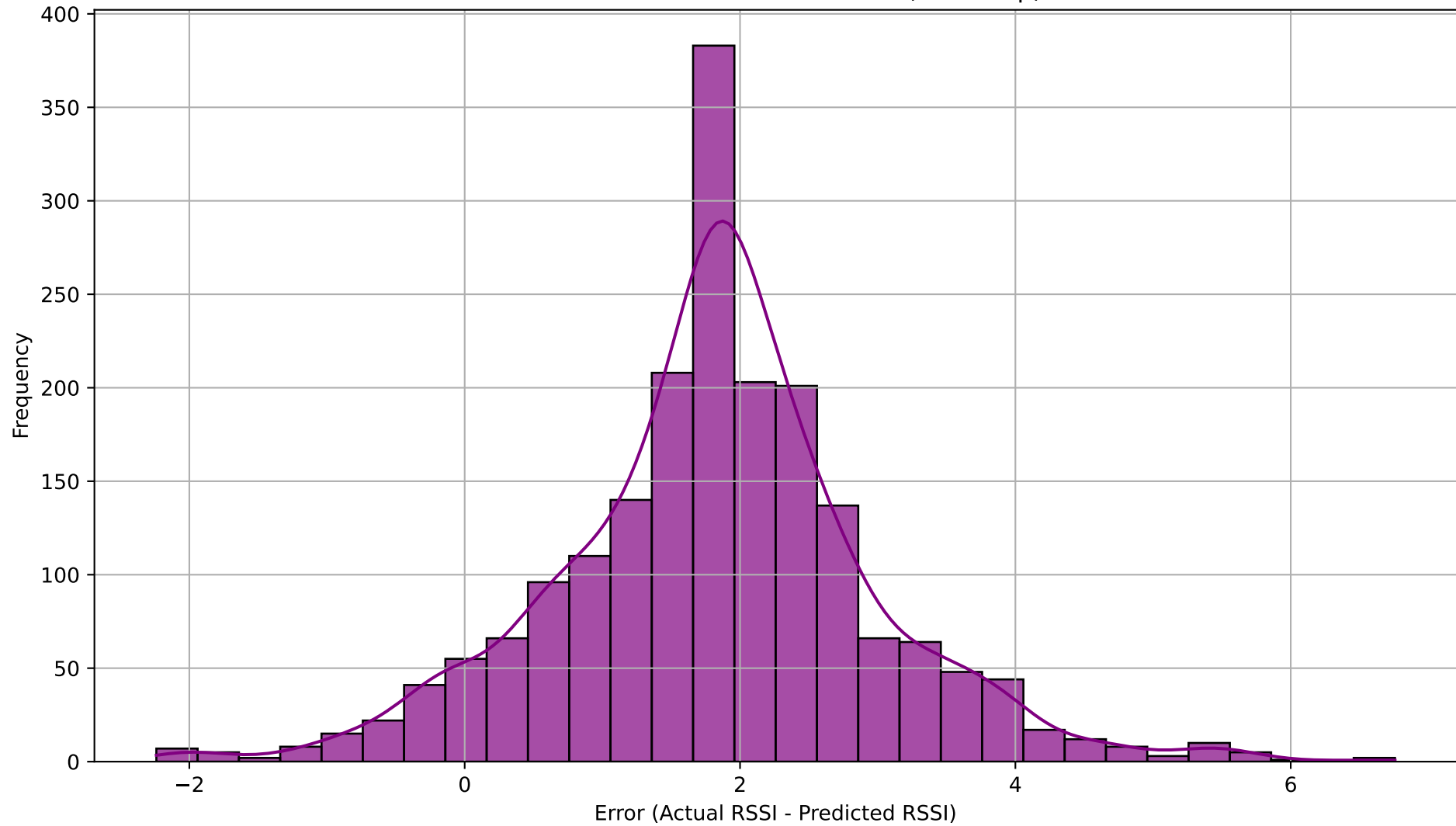
Actual vs Predicted RSSI (First Step)



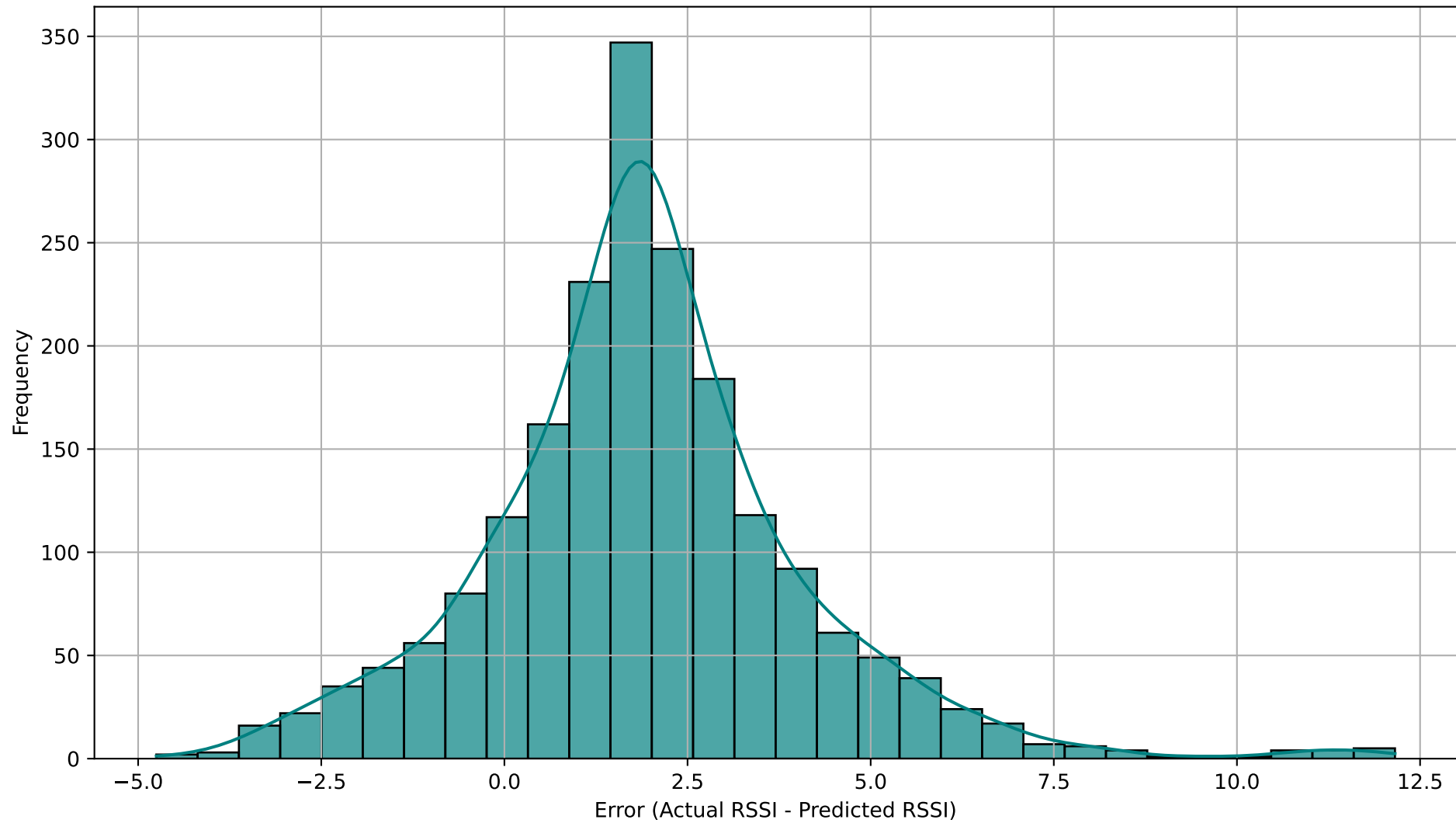
Actual vs Predicted RSSI (Last Step - step 20)



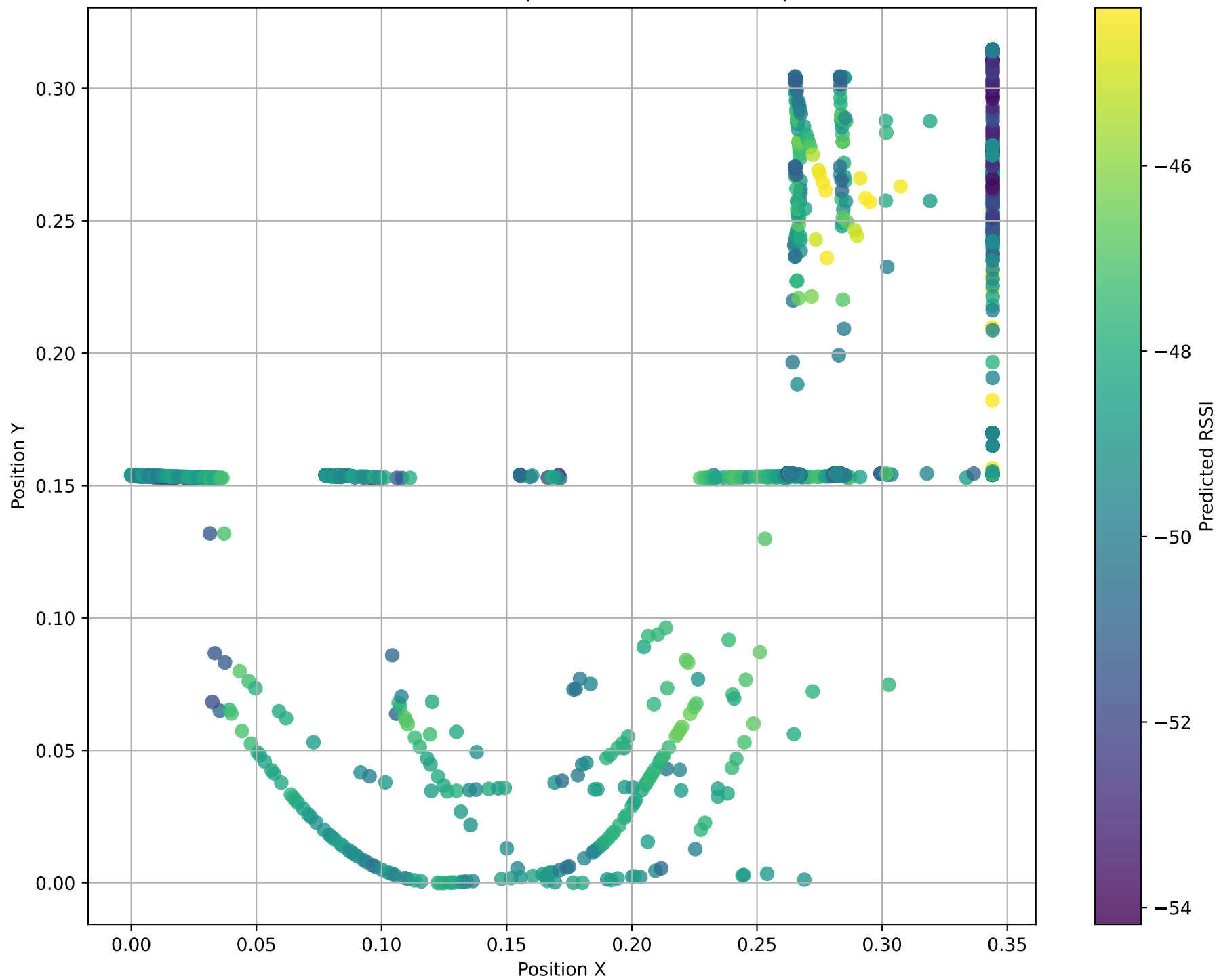
Distribution of Prediction Errors (First Step)



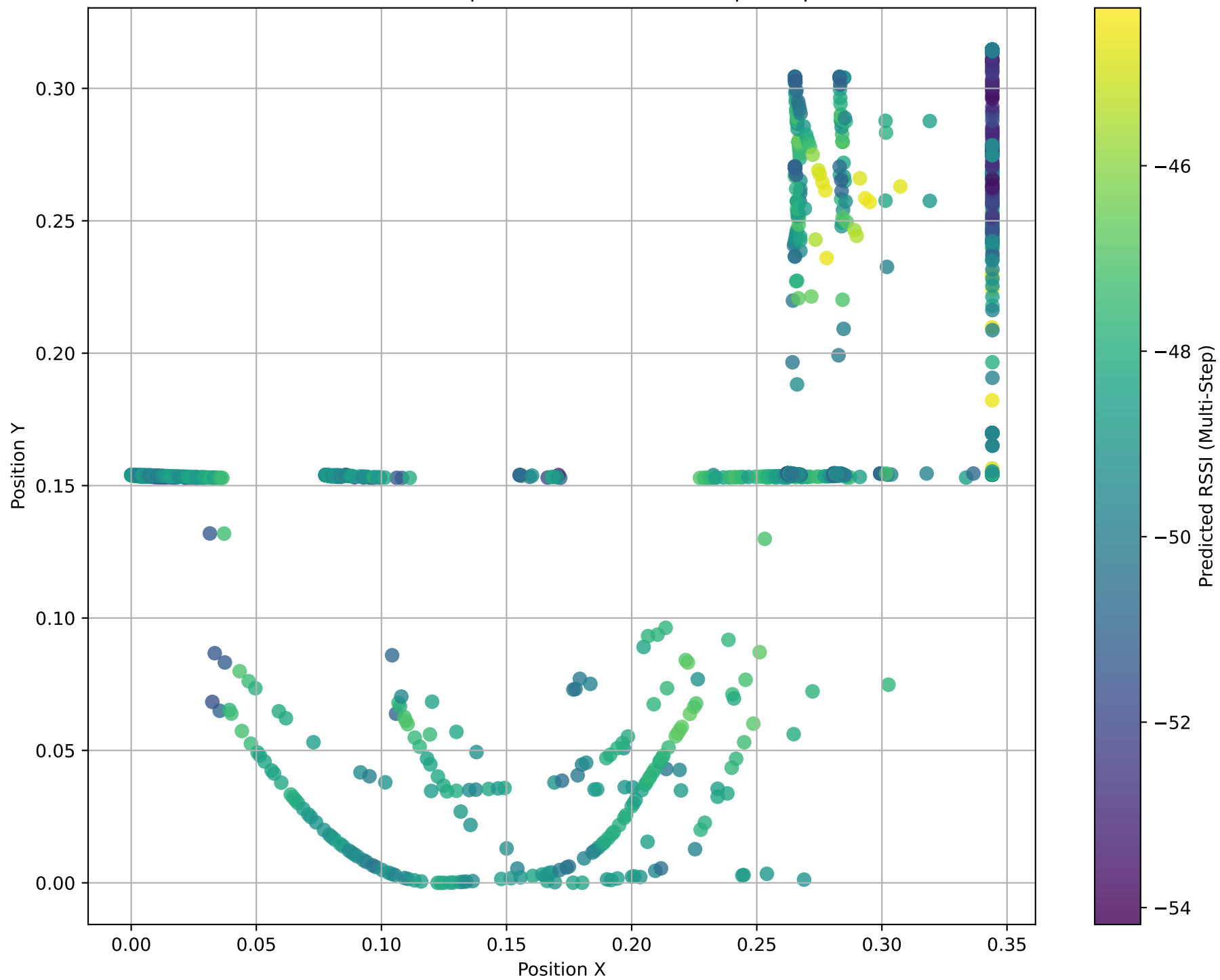
Distribution of Prediction Errors (Last Step - step 20)



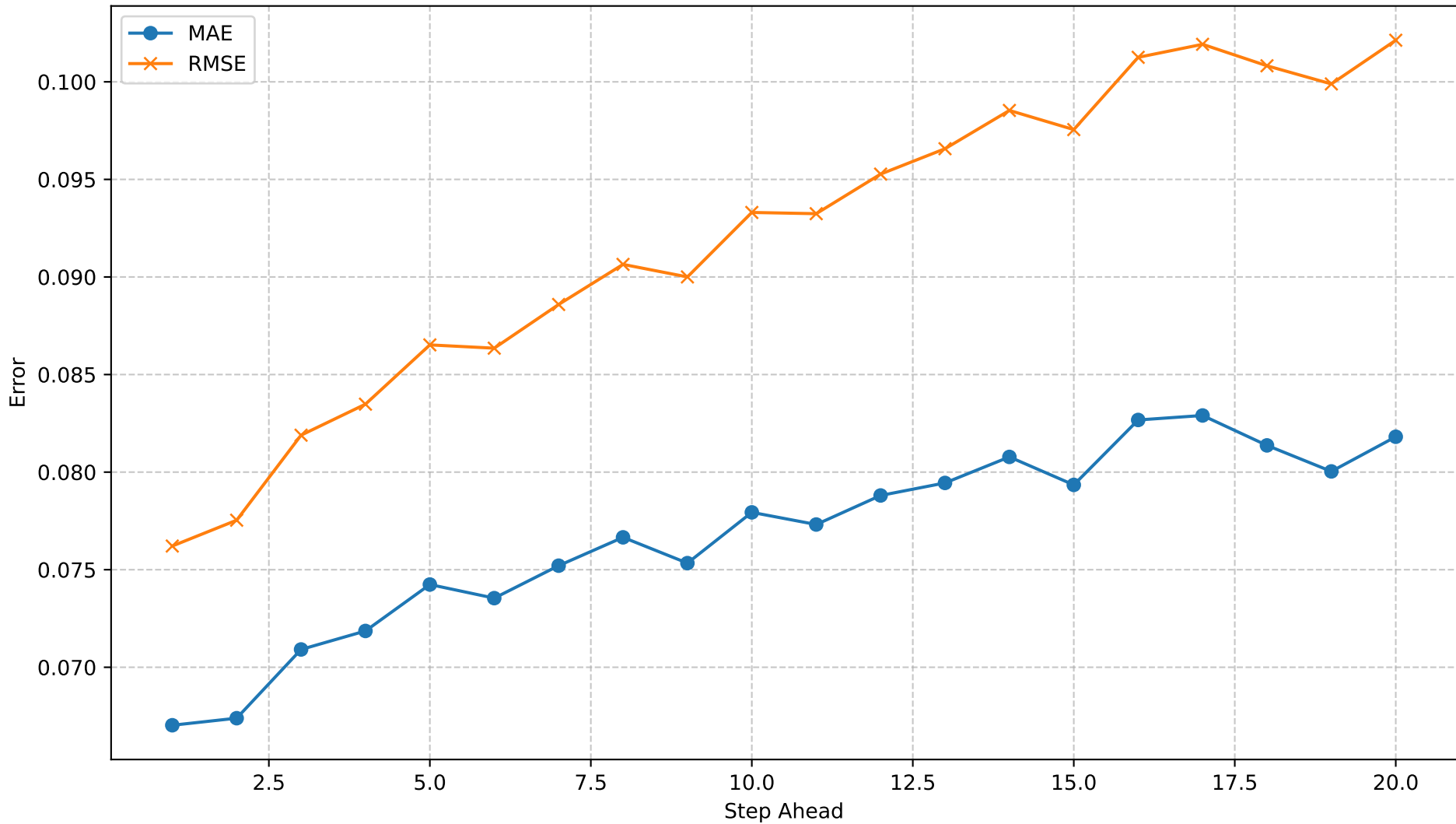
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



Resumo Automático

Dataset: rssi_odom_dataset_11apr25_ajust.parquet

Modelo: LSTM

Forecast steps: 20

MAE / RMSE (dB)

- 1º passo : 0.07 / 0.08
- Mediana : 0.08 / 0.09
- 20º passo : 0.08 / 0.10

Crescimento do erro: 22.1% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 17 (0.08), 16 (0.08), 20 (0.08)

Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.